Assignment 1: Lumen Beauty and BakeBox

New Media Technology and Analytics | data, variables, and association rules

Two stores, two questions. For Lumen Beauty I rank the September livestreams and the eight products, then ask whether the streams covered their own cost. For BakeBox I mine the August baskets and ask which products actually belong together, on which channel.

One cleaning step sits under the Lumen numbers. The session export listed S07 twice. I kept one copy, so September is 12 sessions, not 13. S09's watch time was stored in seconds; I turned it back into minutes. S05 is missing its watch time, and I left it missing. None of the three Lumen tasks uses watch time.

Task 1.1 Two ways to rank a stream

Question. If I rank the 12 sessions by GMV per peak viewer, is the winner also the session with the most viewers? GMV per viewer is session GMV divided by peak viewers.

S05 is first, at 18.08 CNY per viewer. It had 379 peak viewers and 6,852.76 CNY of GMV. S12 is second, at 18.03 CNY per viewer. S12 had 1,427 peak viewers, the largest audience of the month, and 25,728.68 CNY of GMV, also the largest. The top session by GMV per viewer is not the top session by peak viewers.

Peak viewers measures how many people showed up at the busiest moment. GMV per viewer measures how much was sold for each person in that audience. They answer different questions, so the rankings split. The ratio hides the size of the room. S05 leads by only 0.05 CNY per viewer, on a small room and a small total.

What to use. Peak viewers is a reach number. It does not say the stream sold well. S11 is the giveaway night from class: 9,500 comments, 344 peak viewers, and the lowest GMV per viewer in the month, 11.88 CNY. GMV per viewer is the better score of selling quality, and it still needs the audience next to it. A good stream is a high GMV per viewer in a large room. That is S12. I would copy S12. I would not copy S05 just because it leads the ratio by 0.05 CNY.

Task 1.2 Month two

Question. Which two products grew fastest from week 1 to week 4, and what should the store do with each one in month two? Growth is week-4 GMV divided by week-1 GMV. These figures are store-wide sales, from the weekly file.

Cloud Cream 50g is first. It went from 15,418.24 CNY in week 1 to 30,296.71 CNY in week 4. Growth is 1.965, a 96.5% increase, and the path rose every week: 15,418, then 18,926, then 22,730, then 30,297. In week 4 it was 42% of store GMV (71,616.61 CNY). At the 199 CNY list price, week 4 is about 152 units.

Action. Before month two, order Cloud Cream for the week-4 pace: about 152 units a week, about 600 units for four weeks. Put it first in every livestream. It is the fastest grower and the biggest seller.

Mist Toner 150ml is second. It went from 2,466.52 CNY to 4,544.28 CNY. Growth is 1.842, an 84.2% increase, and it also rose every week: 2,467, then 2,924, then 3,451, then 4,544. It is still a small product: 6.3% of week-4 sales. At 89 CNY, week 4 is about 51 units.

Action. Give Mist Toner one fixed pitch in each livestream, as the cheaper step in front of Cloud Cream, and bring in about 50 units a week. Do not stock it like the hero product. In month two, watch whether weekly GMV stays near the week-4 figure of 4,544 CNY.

Task 1.3 Did the streams pay for themselves?

The cost assumption comes first. One hour costs 800 CNY for the host, the venue, and samples. That rate does not include the cost of the products sold. GMV is the value of goods sold, not the cash left after product cost and platform fees.

The 12 sessions run 1,380 minutes, which is 23.0 hours. The streaming bill is 23 x 800 = 18,400 CNY. Stream GMV, the sum of the 12 sessions, is 182,907.21 CNY. The weekly file is a different total: store-wide GMV of 219,383.72 CNY. Streams are 83% of that store total. I compare the bill with stream GMV, not with the store total.

182,907 is 9.9 times 18,400. That multiple is sales over the studio bill. It is not profit of 164,507 CNY.

The conclusion moves with the margin: the share of GMV the store keeps after product cost and fees.

- At a 30% margin the streams contribute about 54,872 CNY. That covers 18,400.
- The break-even margin is 18,400 / 182,907.21 = 10.06%.
- At an 8% margin the streams contribute about 14,633 CNY. That does not cover 18,400.

The streams paid for the host, the venue, and the samples if the store keeps more than about 10% of stream GMV. They did not if the margin is thinner than that. This is the assumption the conclusion stands or falls with. A higher hourly rate is less dangerous: at 1,600 CNY per hour the bill is 36,800 CNY, and a 30% margin still covers it. An 8% margin fails even at the original 800.

Task 2.1 The strongest rules

Question. On all 2,800 August baskets, with minimum support 0.05, which three rules have the highest lift, and what should the merchant do with one of them?

Support is how often the items show up together. Confidence is how often the right-hand product is bought once the left-hand products are already in the basket. Lift compares that with chance. Lift near 1 means no real link. An itemset has to appear in at least 140 orders (5% of 2,800) to be kept.

The three strongest rules are one kit, the apron, the mixing bowl, and the spatula set.

- apron -> mixing bowl + spatula set. Support 0.079, confidence 0.804, lift 10.091.
- mixing bowl + spatula set -> apron. Support 0.079, confidence 0.991, lift 10.091.
- mixing bowl -> apron + spatula set. Support 0.079, confidence 0.759, lift 9.622.

The rule to act on is the second one. Support 0.079 is about 221 of the 2,800 orders. When the mixing bowl and the spatula set are already in the basket, the customer also buys the apron 99.1% of the time. That is about 10 times chance (lift 10.091).

What to do. Sell the apron, the mixing bowl, and the spatula set as one bundle, on the product page and on the live show. When a cart already has two of the three, offer the third. Keep the set intact when you discount. Customers are already buying the three together.

Task 2.2 The bestseller that goes with nothing

Paper doilies are the most common product. They are in 784 of 2,800 orders, support 0.280.

I computed the lift of doilies with every other product from the counts, the same check as in the Week 3 notebook. The lifts run from 0.804 (sugar) to 1.250 (cupcake cases). All of them sit close to 1. At minimum support 0.05, the mining returns no rule that contains doilies. The largest doily pair is 133 orders, under the 140-order line.

High support means the product is common. It does not mean the product predicts the next one. A cross-sell anchor needs a confidence well above the base rate, which shows up as a lift clearly above 1. Doilies fail that test. A box that says "people who bought doilies also bought this" would only repeat the store's ordinary popular products.

Keep or delist should rest on the product's own profit: units sold, the margin on each unit, and the cost of storing and packing it. Doilies are in 28% of orders, so they are a volume item. Keep them if that volume makes money. Drop them if they lose money, even though they lead the store on how often they are bought. Association rules do not make that call.

Task 2.3 One store, two channels

I ran the same mining again, still at minimum support 0.05, on the 816 livestream orders and on the 1,984 storefront orders. The 5% line is 5% of that channel.

Spatula set -> apron is strong on the livestream and absent on the storefront. On the livestream: support 0.271, confidence 1.000, lift 3.692. Every live order that contains the spatula set also contains the apron, and the pair is in 27.1% of live orders. On the storefront the pair does not reach support 0.05, so the rule is absent.

The host pushes this set on the show. A confidence of 1.0 is the script. On the storefront nobody makes that

pitch, so the pair is too rare for the same 5% line. The blended file still shows the apron, bowl, and spatula kit at a lift near 10, because the live orders are mixed into it. A founder who reads only the blended list will think the whole store buys this kit on its own.

Read one rule set per channel.

Storefront bundle page. Build bundles from the storefront rules. Two strong ones there: piping bag -> nozzle set (support 0.109, confidence 1.000, lift 9.185) and bamboo whisk -> matcha powder (support 0.100, confidence 1.000, lift 8.371). Leave the apron kit off the hero slot of that page.

Live show. Keep offering the spatula set with the apron and the mixing bowl. In that room the confidence is 1.0. The host can also mention the baking pairs, and the script should follow the live rules rather than the storefront page.

AI-use appendix

Cursor's AI assistant (Grok) helped me write the notebook code and draft this report. The code follows the Week 2 and Week 3 class notebooks: drop the duplicate session, fix the watch-time unit, compute GMV per viewer, compute week-4 GMV over week-1 GMV, and mine rules with mlxtend at min_support = 0.05 with no confidence filter. I kept those choices. Every figure in this report is printed by assignment_1.ipynb.